

Lecture 1 — Introduction to Mathematical Modelling & Kinematics of Particles & Newton's Laws

MATH3001(5005) — Applied Mathematical Modelling (Topics)

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Welcome

Goals today:

- Unit information
- Intro to Mathematical Modelling
- Kinematics of Particles
- Newton's Laws and Applications



MATH3001(5005) Applied Mathematical Modelling (Topics)

- Unit Coordinator:
 - Prof. Benchawan Wiwatanapataphee
- Lecturer-in-charge:
 - Prof. Benchawan Wiwatanapataphee, office: 314.347C, email: b.wiwatanapataphee@curtin.edu.au
 - Prof. Yong Hong Wu, office: 314.363, email: Y.Wu@curtin.edu.au
- Weekly Contact Hours
 - Lecture: 2 hours
 - Tutorial: 1 hour
 - Workshop: 1 hour



Learning Outcomes

By the end of this unit, you should be able to

- Apply Newton's Laws to mathematical modelling
- Develop continuum mechanics models
- Solve initial and boundary value problems
- Analyse DE/PDE models
- Apply mathematics to engineering, biology and science
- Interpret computational and analytical results



Weekly Topics

Lecture Week	Topic
1	Introduction & Newton's Laws
2	Vectors and Tensors
3	Stress Tensor
4	Constitutive Relations
5	Heat Transfer
6	Fluid Flow
8	PDE Models
9	Geomechanics
10	Water Filtration
11	Traffic Flow



Lecture Week Topic

12	Integrated Modelling Project
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Assessment

Assessment	Weight
Mid-semester Test	30%
Group Project Assignment	20%
Final Examination	50%

To pass the unit:

- Final mark $\geq 50\%$



Group Project

Worth 20%

Students work in teams of **up to 3 students**

Project objectives

- Solve a real-world mathematical modelling problem
- Apply concepts learned throughout the semester
- Develop teamwork and project management skills
- Present mathematical results clearly



Project Learning Goals

The project is designed to help you

- Connect mathematics with real applications
- Develop modelling skills
- Learn computational implementation
- Interpret numerical results
- Improve scientific communication
- Work effectively in teams



Possible Project Topics

Examples include

- Heat conduction
- Diffusion models
- Traffic flow
- Water filtration
- Population dynamics
- Mechanical vibrations
- Fluid transport
- Environmental modelling

Students may also propose their own project (subject to approval).



What Your Team Should Produce

Your submission should include

- Problem statement
- Mathematical model
- Assumptions
- Governing equations
- Solution method
- Numerical simulation
- Interpretation of results
- Conclusions



Good Teamwork

Successful teams

- Meet regularly
- Divide responsibilities fairly
- Keep shared notes
- Use GitHub or OneDrive
- Communicate frequently
- Respect deadlines

Remember:

Every member should understand the entire project.



Suggested Timeline

Week	Task
3	Form teams
4	Choose topic
5	Project proposal
6-9	Model development
10	Simulation & analysis
11	Writing report
12	Submit assignment
13	Presentation



Assessment Criteria

Your project will be evaluated on

- Mathematical modelling
- Correctness
- Mathematical analysis
- Numerical implementation
- Interpretation
- Report quality
- Team contribution



Academic Integrity

You are encouraged to

- Discuss ideas
- Work collaboratively within your group

You must not

- Copy another group's work
- Use AI without permission
- Submit work that is not your own

Always acknowledge any external resources.



Tips for Success

- Start early.
- Ask questions.
- Attend workshops.
- Practice modelling every week.
- Keep your mathematics organised.
- Use MATLAB or Python effectively.



Communication

Questions?

- Blackboard Discussion Board
- Consultation Hours (Monday 2–3pm, Thursday 10–11am)
- Email the Unit Coordinator (b.wiwatanapataphee@curtin.edu.au)

We are here to help you succeed.



Introduction to Mathematical Modelling



Definition of Mathematical Modelling

Mathematical modelling represents real-world systems using mathematics to describe their essential behaviour, enabling analysis, understanding and prediction.

- A mathematical model is a simplified representation of a real-world system.
- It includes only the features relevant to the problem being studied.

Applications:

- population growth
- heat transfer
- traffic flow
- fluid dynamics
- disease transmission
- water filtration



- structural mechanics.



The Mathematical Modelling Process

 A mathematical model translates a real-world problem into mathematics so that it can be analysed and used for prediction.

1. Define the problem

- Identify the question and objectives.
- Determine the system boundaries.

2. Make assumptions

- Simplify reality while retaining the essential features.
- State assumptions clearly.

3. Formulate the model

- Identify governing physical laws (e.g., conservation laws, Newton's laws).
- Define variables and parameters.
- Derive the mathematical equations.



1. Solve the model

- Obtain analytical or numerical solutions.
- Verify that the solution is mathematically consistent.

2. Interpret and validate

- Compare predictions with observations or experimental data.
- Assess whether assumptions are reasonable.
- Refine the model if necessary.



Modelling is an iterative process: if predictions do not agree with observations, revise the assumptions or the model and repeat the cycle.

Mathematical Models of Real-World Systems

- Real-world systems are represented by different mathematical models depending on the underlying physical mechanisms.
- The choice of model determines how the system is analysed, simulated, and used for prediction.

Examples include:

System	Mathematical Model
Population Growth	Differential Equation
Traffic Flow	Conservation Law
Heat Conduction	Heat Equation
Fluid Flow	Navier–Stokes Equations
Disease Spread	SIR Model
Water Filtration	Advection–Diffusion Equation



Variables, Parameters and State Variables

A mathematical model typically contains:

- **Independent variables:** quantities that describe the domain of the problem (e.g. time, space).
- **Dependent variables:** quantities whose values depend on the independent variables.
- **Parameters:** fixed quantities that characterise the system.
- **State variables:** variables describing the current state of the system.



A mathematical model establishes relationships among these quantities through equations derived from physical laws, empirical observations, or fundamental principles.

The objective is to predict how the **state variables** change as the **independent variables** vary, for a given set of **parameters**.





Example: Newton's Cooling Law

$$\frac{dT}{dt} = -k(T - T_a),$$

- t is time (**independent variable**),
- $T(t)$ is the temperature of the object at time t (**dependent/state variable**),
- T_a is the ambient temperature (**parameter**),
- k is a positive constant that depends on the properties of the object and the surrounding environment (**parameter**).



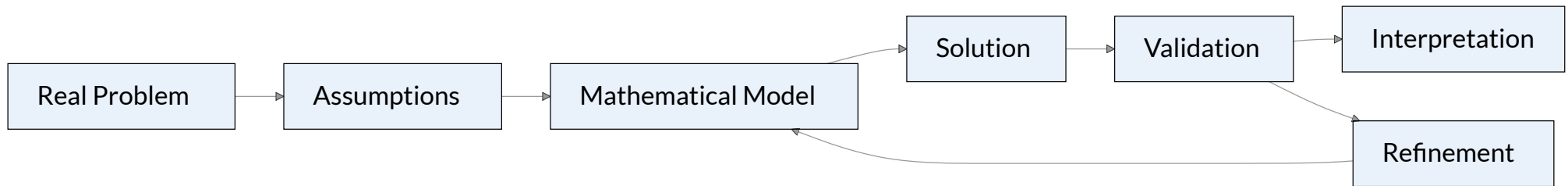
Why Build Mathematical Models?

- Mathematical models are used to
 - Understand physical mechanisms.
 - Predict future behaviour.
 - Test scientific hypotheses.
 - Design engineering systems.
 - Optimise performance.
 - Reduce experimental costs.
- Examples include
 - Predicting temperature distributions in a reactor.
 - Estimating contaminant transport in groundwater.
 - Forecasting traffic congestion.
 - Designing aircraft wings.



The Modelling Cycle

Mathematical modelling is an iterative process rather than a single calculation. The modelling cycle consists of several stages.



The modelling cycle may require several iterations before the mathematical model adequately represents the real system.





Example: Cooling Coffee

Problem: Determine how the temperature of a cup of coffee changes over time.

- **Assumptions:** Coffee is perfectly mixed. Ambient temperature remains constant. Heat loss is proportional to the temperature difference.
- **Model Formulation:** $\frac{dT}{dt} = -k(T - T_a)$,
 - $T(t)$ is the coffee temperature,
 - T_a is the ambient temperature,
 - $k > 0$ is a cooling constant.
- **Solution:** $T(t) = T_a + (T_0 - T_a)e^{-kt}$.



Physical Systems



A **physical system** is the portion of the universe selected for investigation.

• Examples

- A moving vehicle,
- A heat exchanger,
- A chemical reactor,
- A river,
- A traffic network,
- A biological cell.



System Boundary

Everything outside the system is called the surroundings or environment. The interaction between the system and its surroundings occurs through its boundary.

- The boundary separates the system from its surroundings.
- Boundaries may be **fixed**, **moving**, **real**, or **imaginary**.
- Across the boundary there may be transfer of **mass**, **momentum**, and **energy**.

 Choosing an appropriate system boundary is one of the most important steps in constructing a mathematical model.

Control Volumes



Many engineering systems involve the flow of fluids through pipes, channels, heat exchangers and reactors. Rather than tracking individual fluid particles, we analyse a fixed region in space known as a control volume. The boundary surrounding this region is called the control surface.

- **Examples:**

- a water tank,
- a pipe section,
- a pump,
- a combustion chamber.



Assumptions and Idealisations



Every mathematical model is a simplification of reality.

To obtain a tractable model, we retain the dominant physical effects and neglect those with little influence on the results.

“All models are wrong, but some are useful.”

— **George E. P. Box**



A good model is not the most detailed model—it is the simplest model that accurately answers the engineering question.



Why make assumptions?

- Simplify complex systems
- Focus on the dominant physics
- Reduce mathematical complexity
- Obtain practical solutions
- Improve computational efficiency

Common assumptions

- Negligible air resistance
- Constant density
- Uniform temperature
- One-dimensional flow
- Steady-state conditions
- Incompressible fluid



- Frictionless surfaces



Dimensional Analysis

Fundamental dimensions include:

Quantity	Dimension
Length	L
Mass	M
Time	T
Temperature	Θ

Example

- Velocity

$$v = \frac{\text{distance}}{\text{time}}, \quad \text{so} \quad [v] = LT^{-1}.$$

- Acceleration

$$a = \frac{\text{velocity}}{\text{time}}, \quad \text{so} \quad [a] = LT^{-2}.$$



Dimensional Consistency

 A physically meaningful equation must be **dimensionally consistent**—every term must have the **same physical dimensions**.

Example

$$s = ut + \frac{1}{2}at^2.$$

DIMENSIONS:

$$[L] = [LT^{-1}][T] + [LT^{-2}][T^2].$$

HENCE,

$$[L] = [L] + [L],$$

which confirms dimensional consistency.



Scaling and Characteristic Quantities

 **Scaling** estimates the relative size of physical quantities and identifies the dominant processes in a system.

Example: Characteristic Timescale of a Vehicle

- Suppose a vehicle travels a characteristic distance (L) at a characteristic speed (U).
- The characteristic time is

$$T_c = \frac{L}{U}.$$



For

$$L = 1000 \text{ m}, \quad U = 25 \text{ m/s},$$

we obtain

$$T_c = \frac{1000}{25} = 40 \text{ s}.$$



Scaling helps engineers:

- estimate the order of magnitude of physical quantities,
- identify the dominant physical mechanisms,
- simplify governing equations, and
- compare different physical systems.



Key idea: Characteristic quantities provide a reference scale for analysing physical systems and form the basis of **non-dimensionalisation**.



2. Kinematics and Newton's Laws



Many physical quantities are described by both **magnitude** and **direction**.
These quantities are represented using **vectors**.

Vector (magnitude and direction)

- Velocity
- Acceleration
- Force
- Momentum
- Heat flux

Scalar (magnitude only)

- Mass
- Temperature
- Time
- Energy



▸ Pressure

 Vector arithmetic provides the mathematical language for expressing **Newton's laws of motion** and many **conservation laws** used in engineering and science.



Basic Vector Arithmetic

i **Scalar:** A quantity with magnitude only. Examples include Mass, Time, Temperature, Energy, Pressure, Density.

i **Vector:** A quantity with both magnitude and direction. Examples include Velocity, Force, Displacement, Acceleration.

- Throughout this unit vectors will be represented by bold symbols, for example $\mathbf{v}$, $\mathbf{F}$, $\mathbf{r}$, $\mathbf{a}$.
- Their magnitudes are denoted by $|\mathbf{v}|$, $|\mathbf{F}|$, $|\mathbf{r}|$, $|\mathbf{a}| \dots$



Geometric Representation of a Vector

- A vector is represented graphically by a directed line segment.
 - The length of the arrow represents the magnitude.
 - The arrowhead indicates the direction.
 - The initial point is called the **tail**.
 - The terminal point is called the **head**.
- Suppose a vector begins at point A and ends at point B . Then
 - the vector is written as $\overrightarrow{AB}$.
 - Its magnitude is $|\overrightarrow{AB}|$.



Cartesian Representation and Vector Operations

Figure 1: Cartesian Representation of a Vector.

- In 3-D Cartesian coordinates, a vector is represented by its components.
- Suppose $A = (x, y, z)$. Then the position vector of A measured from the origin is $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k} \equiv (x, y, z)$ where $\mathbf{i}, \mathbf{j}, \mathbf{k}$ are the unit vectors in the x, y, z directions respectively.



Magnitude of a Vector

The length (or magnitude) of a vector $\mathbf{a} = (a_x, a_y, a_z)$ is

$$|\mathbf{a}| = \sqrt{a_x^2 + a_y^2 + a_z^2}.$$



Example

Find the magnitude of the vector $\mathbf{a} = (3, 4, 5)$.

$$|\mathbf{a}| = \sqrt{3^2 + 4^2 + 5^2} = \sqrt{9 + 16 + 25} = \sqrt{50} = 5\sqrt{2}.$$



Equality of Vectors

Two vectors are equal if they have the **same magnitude**, and the **same direction**, regardless of where they are located.



Example

$$\mathbf{a} = (1, 2, 3), \quad \mathbf{b} = (1, 2, 3) \implies \mathbf{a} = \mathbf{b}.$$

even if the vectors originate from different points.



Vector Addition

If $\mathbf{u} = (u_1, u_2, u_3)$, and $\mathbf{v} = (v_1, v_2, v_3)$, then

$$\mathbf{u} + \mathbf{v} = (u_1 + v_1, u_2 + v_2, u_3 + v_3).$$

Vector addition is performed by adding the corresponding components.



Example

Let

$$\mathbf{u} = (1, 0), \quad \mathbf{v} = (1, 1).$$

Then

$$\mathbf{u} + \mathbf{v} = (1 + 1, 0 + 1) = (2, 1).$$



Geometric Interpretation

Vector addition can be visualised using either the **parallelogram law** or the **triangle law**.

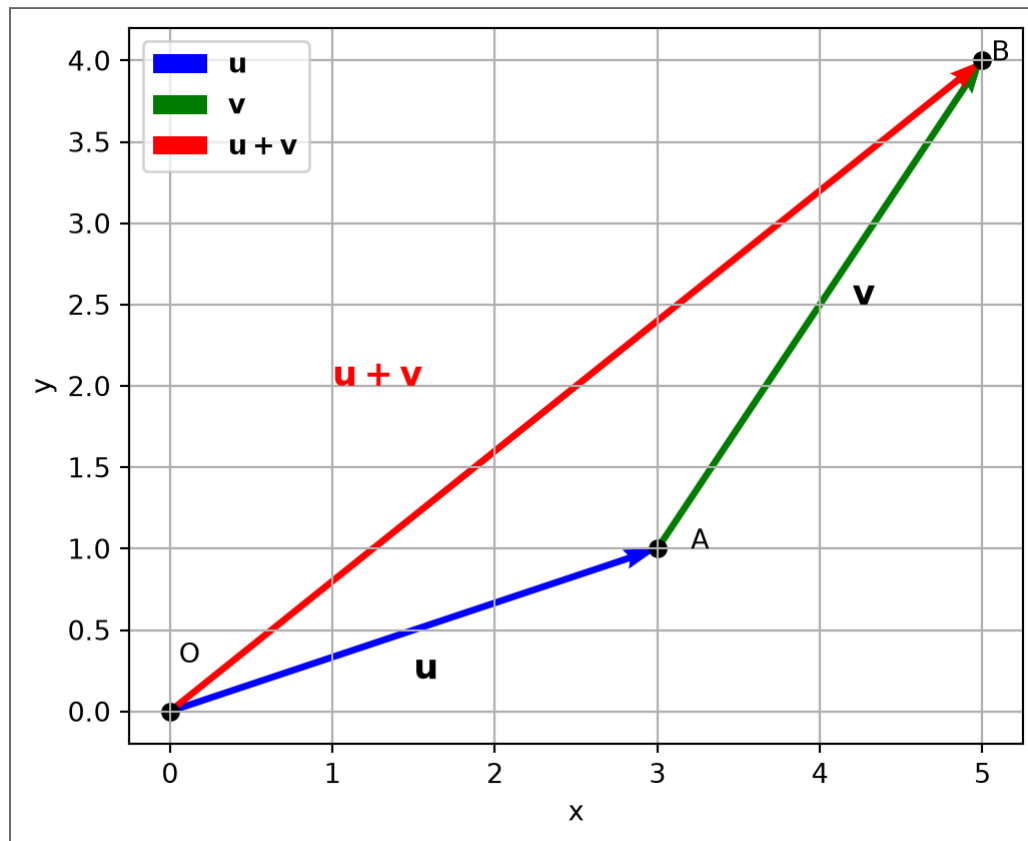


Figure 2: Vector Addition Using the Triangle Law



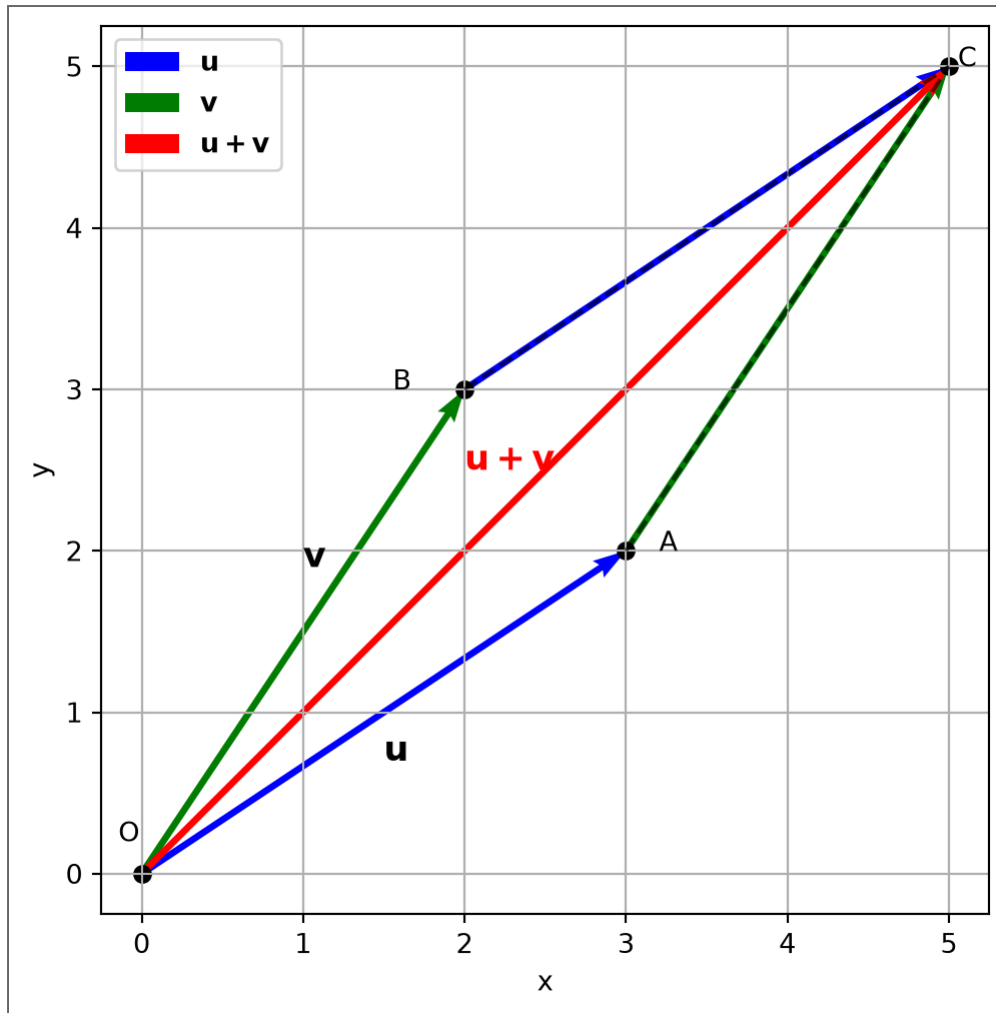


Figure 3: Vector Addition Using the Parallelogram Law



Parallelogram Law: To add two vectors, place them so their tails coincide. The resultant vector is the diagonal of the parallelogram formed by the two vectors.



Vector Subtraction

Vector subtraction is defined as

$$\mathbf{u} - \mathbf{v} = \mathbf{u} + (-\mathbf{v}).$$

The vector $-\mathbf{v}$ is the vector with the same magnitude as $\mathbf{v}$ but in the opposite direction.



Example

Let $\mathbf{u} = (5, 4)$, $\mathbf{v} = (2, 1)$. Then

$$\mathbf{u} - \mathbf{v} = (5 - 2, 4 - 1) = (3, 3).$$



Scalar Multiplication

Multiplying a vector by a scalar changes its magnitude. If $\lambda \in \mathbb{R}$, then

$$\lambda \mathbf{a} = (\lambda a_x, \lambda a_y, \lambda a_z).$$

- If $\lambda > 1$, the vector becomes longer.
- If $0 < \lambda < 1$, the vector becomes shorter.
- If $\lambda < 0$, the direction is reversed.

Example

Let $\mathbf{a} = (2, -1)$. Then

$$2\mathbf{a} = (4, -2), \quad 0.5\mathbf{a} = (1, -0.5), \quad -1\mathbf{a} = (-2, 1).$$



Unit Vectors

- A unit vector has magnitude equal to one. The standard Cartesian unit vectors are $\mathbf{i} = (1, 0, 0)$, $\mathbf{j} = (0, 1, 0)$, $\mathbf{k} = (0, 0, 1)$.

- Any vector

$$\mathbf{a} = (a_x, a_y, a_z)$$

may be written as

$$\mathbf{a} = a_x \mathbf{i} + a_y \mathbf{j} + a_z \mathbf{k}.$$

The unit vector in the direction of $\mathbf{a}$ is

$$\hat{\mathbf{a}} = \frac{\mathbf{a}}{|\mathbf{a}|}.$$





Example

Let $\mathbf{a} = (3, 4)$. Since $|\mathbf{a}| = 5$, the unit vector is

$$\hat{\mathbf{a}} = \frac{1}{5}(3, 4) = \left(\frac{3}{5}, \frac{4}{5}\right).$$



POSITION VECTORS

- The position of a particle relative to the chosen origin is described by the **position vector**.
- Suppose a particle is located at $P(x, y, z)$. Then the position vector of $\mathbf{r}(t)$ is $\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}$.
Alternatively,
 $\mathbf{r}(t) = (x(t), y(t), z(t))$.



Example

If $P = (3, 4, 2)$, then the position vector is

$$\mathbf{r} = 3\mathbf{i} + 4\mathbf{j} + 2\mathbf{k} = (3, 4, 2).$$

Its distance from the origin is $|\mathbf{r}| = \sqrt{3^2 + 4^2 + 2^2} = \sqrt{29}$.



DISPLACEMENT VECTORS

- A particle moves from $P_1(x_1, y_1, z_1)$ to $P_2(x_2, y_2, z_2)$. Then $\Delta \mathbf{r} = \mathbf{r}_2 - \mathbf{r}_1 = (x_2 - x_1)\mathbf{i} + (y_2 - y_1)\mathbf{j} + (z_2 - z_1)\mathbf{k}$.

EXAMPLE

A particle moves from $P_1 = (2, 1, 3)$ to $P_2 = (7, 5, 1)$.

The **displacement vector** is

$$\begin{aligned}\Delta \mathbf{r} &= \mathbf{r}_2 - \mathbf{r}_1 \\ &= (7 - 2)\mathbf{i} + (5 - 1)\mathbf{j} + (1 - 3)\mathbf{k} \\ &= 5\mathbf{i} + 4\mathbf{j} - 2\mathbf{k}.\end{aligned}$$

The **magnitude** of the displacement is

$$\begin{aligned}|\Delta \mathbf{r}| &= \sqrt{5^2 + 4^2 + (-2)^2} \\ &= \sqrt{45} \\ &= 3\sqrt{5} \approx 6.71.\end{aligned}$$

 Displacement is the straight-line change in position; distance travelled is the length of the path.



VELOCITY VECTORS

- Velocity describes how rapidly the position of a particle changes with time.
- If the particle has position vector $\mathbf{r}(t)$, its velocity is defined as

$$\mathbf{v} = \frac{d\mathbf{r}}{dt} = \dot{x}\mathbf{i} + \dot{y}\mathbf{j} + \dot{z}\mathbf{k} \quad \text{or} \quad \mathbf{v} = (\dot{x}, \dot{y}, \dot{z}).$$

- The magnitude of the velocity, $|\mathbf{v}|$, is called the **speed**.

Example

- Let $\mathbf{r}(t) = (t, t^2, t^3)$. Then

$$\mathbf{v}(t) = \frac{d\mathbf{r}}{dt} = (1, 2t, 3t^2).$$

The speed is

$$|\mathbf{v}(t)| = \sqrt{1 + (2t)^2 + (3t^2)^2} = \sqrt{1 + 4t^2 + 9t^4}.$$



ACCELERATION VECTORS

Acceleration describes how rapidly the velocity of a particle changes with time.

If the particle has velocity $\mathbf{v}(t)$, its acceleration is defined as

$$\mathbf{a} = \frac{d\mathbf{v}}{dt} = \frac{d^2\mathbf{r}}{dt^2}.$$



Example

Let $\mathbf{r}(t) = (t, t^2, t^3)$. Then,

$$\mathbf{a}(t) = \frac{d\mathbf{v}}{dt} = (0, 2, 6t).$$



PARTICLE MOTION ON A HYPERBOLA

 Describe the path of the particle, its velocity and its acceleration if a particle has position vector $\mathbf{r}(t) = (a \cosh(\omega t), b \sinh(\omega t), 0)$, $0 \leq t < \infty$, where a , b and ω are positive constants.

The particle coordinates are

$$x = a \cosh(\omega t), \quad y = b \sinh(\omega t), \quad z = 0.$$

Since

$$\cosh^2(\omega t) - \sinh^2(\omega t) = 1,$$

we obtain

$$\left(\frac{x}{a}\right)^2 - \left(\frac{y}{b}\right)^2 = 1.$$



Hence, the particle moves along the hyperbola

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1,$$

which lies in the xy -plane and is centred at the origin.

For a left branch hyperbola, we can use the parametrization

$$\mathbf{r}(t) = (-a \cosh(\omega t), b \sinh(\omega t), 0).$$

If you want it shifted left/right, use

$$\mathbf{r}(t) = (h - a \cosh(\omega t), b \sinh(\omega t), 0),$$

where $(h, 0)$ is the hyperbola center.

The velocity is obtained by differentiating the position vector.

$$\mathbf{v}(t) = \frac{d\mathbf{r}}{dt} = (a\omega \sinh(\omega t), b\omega \cosh(\omega t), 0).$$



Equivalently,

$$\mathbf{v}(t) = (a\omega \sinh(\omega t), b\omega \cosh(\omega t), 0).$$

The speed is

$$|\mathbf{v}(t)| = \sqrt{(a\omega \sinh(\omega t))^2 + (b\omega \cosh(\omega t))^2}.$$

Differentiating the velocity vector gives the acceleration.

$$\mathbf{a}(t) = \frac{d\mathbf{v}}{dt} = (a\omega^2 \cosh(\omega t), b\omega^2 \sinh(\omega t), 0).$$



In a pursuit problem, one object is constrained to follow another object in a specified manner.

EXAMPLE

- A comet is attracted to a planet and moves with constant speed, always pointing towards the planet. The path of the comet is called a pursuit curve.
- Let the comet move along a known path
$$\mathbf{r}_A(t) = (x_A(t), y_A(t), z_A(t)),$$
- Assume that the comet travels with constant speed v .
- Since the velocity of the comet is always directed from the comet towards the planet, the direction of motion is parallel to
$$\mathbf{r}_A(t) - \mathbf{r}(t).$$



Hence,

$$\frac{d\mathbf{r}}{dt} = v \frac{\mathbf{r}_A(t) - \mathbf{r}(t)}{|\mathbf{r}_A(t) - \mathbf{r}(t)|}.$$

- This is the governing equation for the pursuit curve.
- In component form, this gives the system

$$\begin{aligned} \frac{dx}{dt} &= v \frac{x_A(t) - x(t)}{\sqrt{(x_A(t) - x(t))^2 + (y_A(t) - y(t))^2 + (z_A(t) - z(t))^2}}, \\ \frac{dy}{dt} &= v \frac{y_A(t) - y(t)}{\sqrt{(x_A(t) - x(t))^2 + (y_A(t) - y(t))^2 + (z_A(t) - z(t))^2}}, \\ \frac{dz}{dt} &= v \frac{z_A(t) - z(t)}{\sqrt{(x_A(t) - x(t))^2 + (y_A(t) - y(t))^2 + (z_A(t) - z(t))^2}}. \end{aligned}$$



Given the initial position of the comet,

$$\mathbf{r}(0) = (x(0), y(0), z(0)).$$

Example (Pursuit Curve)

Suppose a comet moves along the path

$$\mathbf{r}_A(t) = (0.4 \cosh t, 0.7 \sinh t, 0), \quad 0 \leq t < \infty,$$

- At initial time $t = 0$, the comet is located at $(0.4, 0, 0)$ and a planet is located at the origin.
- The comet moves with constant speed $v = 1$ and its direction of motion is always pointed towards the planet.



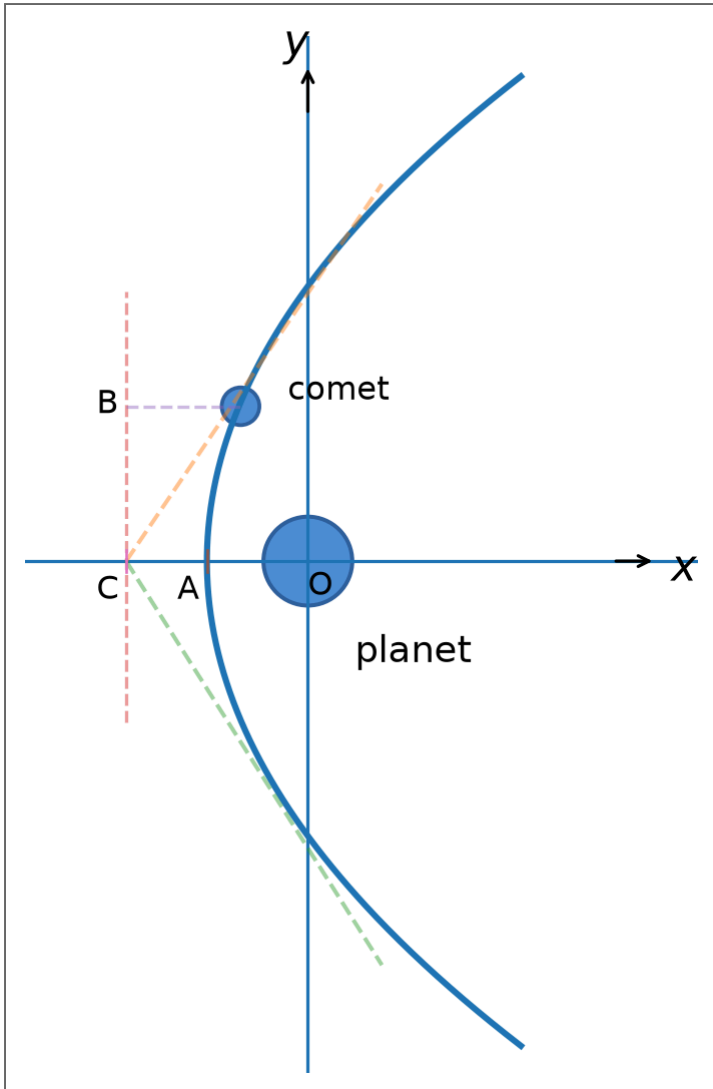


Figure 4: Pursuit Curve



- The governing equations for the pursuit curve are

$$\frac{dx}{dt} = \frac{0.4 \cosh t - x}{\sqrt{(0.4 \cosh t - x)^2 + (0.7 \sinh t - y)^2}},$$

$$\frac{dy}{dt} = \frac{0.7 \sinh t - y}{\sqrt{(0.4 \cosh t - x)^2 + (0.7 \sinh t - y)^2}}.$$

- The pursuit curve in **Figure 4** is obtained by solving this system of differential equations with initial conditions numerically.
- The comet moves with constant speed and its direction of motion is always pointed towards the planet, where $AC = a$ and $BC = b$.



Notes:

The following concepts form the foundation of particle kinematics.

Quantity	Mathematical Definition
Position	$\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$
Displacement	$\Delta\mathbf{r} = \mathbf{r}_2 - \mathbf{r}_1$
Velocity	$\mathbf{v} = \frac{d\mathbf{r}}{dt}$
Acceleration	$\mathbf{a} = \frac{d\mathbf{v}}{dt} = \frac{d^2\mathbf{r}}{dt^2}$

These quantities provide the mathematical description of particle motion and form the basis for Newton's laws of motion, which relate the motion of a particle to the forces acting upon it.



3. Newton's Laws of Motion and Applications

Newton's Laws of Motion

- In the seventeenth century, **Sir Isaac Newton** formulated three fundamental laws that describe the motion of particles under the action of forces.
- These laws form the foundation of classical mechanics and provide the basis for many mathematical models in engineering and science.
- By combining Newton's laws with appropriate force models, we can derive **differential equations** governing the motion of particles, rigid bodies and continua.



- Let us first define two important physical quantities.

 **Mass** is a measure of the amount of matter contained in a body. It is defined by $m = \rho V$, where m is the mass (kg), ρ is the density (kg/m^3), V is the volume (m^3).

 **Momentum** is a measure of the motion of a body. The linear momentum of a particle is defined as the product of its mass and velocity, It is defined by $\mathbf{p} = m\mathbf{v}$, where $\mathbf{p}$ is the momentum ($\text{kg}\cdot\text{m/s}$), m is the mass (kg), and $\mathbf{v}$ is the velocity (m/s).



Newton's First Law (Law of Inertia)

- A particle remains at rest or continues to move with constant velocity in a straight line unless acted upon by a resultant external force.
- This law introduces the concept of inertia.

! Newton's First Law

If the resultant force is zero,

$$\sum \mathbf{F} = \mathbf{0}, \quad \text{then} \quad \mathbf{a} = \mathbf{0}.$$

Consequently, the velocity of the particle remains constant.

Example

- A hockey puck sliding on a perfectly frictionless surface continues moving at constant speed until another force acts on it.



Newton's Second Law

The rate of change of momentum of a particle is equal to the resultant external force acting on the particle.

Mathematically,

$$\sum \mathbf{F} = \frac{d\mathbf{p}}{dt}.$$

For constant mass, $\mathbf{p} = m\mathbf{v}$.

! Newton's Second Law

$$\sum \mathbf{F} = m \frac{d\mathbf{v}}{dt} = m\mathbf{a}.$$

- This is the fundamental equation used throughout engineering mechanics.



The equation may also be written as

$$m \frac{d^2 \mathbf{r}}{dt^2} = \sum \mathbf{F}.$$

- Hence, Newton's Second Law naturally leads to a second-order ordinary differential equation for the particle's position.

Throughout this unit we shall use the **SI system of units**.

Quantity	SI Unit
Length	metre (m)
Mass	kilogram (kg)
Time	second (s)
Force	newton (N)
Velocity	m/s
Acceleration	m/s ²



Example

- A force of 20 N acts on a particle of mass 5 kg.
- Using Newton's Second Law,

$$\mathbf{a} = \frac{\mathbf{F}}{m}.$$

- The acceleration is

$$a = \frac{20}{5} = 4 \text{ m/s}^2.$$



Newton's Third Law

- For every action there is an equal and opposite reaction.
- Whenever two bodies interact, the force exerted by the first body on the second is equal in magnitude and opposite in direction to the force exerted by the second body on the first.
- If body A exerts a force on body B , $\mathbf{F}_{AB}$,
- then body B exerts a force on body A , $\mathbf{F}_{BA}$.

! Newton's Third Law

$$\mathbf{F}_{BA} = -\mathbf{F}_{AB}.$$

- These forces always occur in pairs and act on different bodies.



NEWTON'S THIRD LAW ON A HORIZONTAL PLANE

Consider a block resting on a horizontal table.

- The block exerts a downward force on the table equal to its weight.
- The table exerts an upward force on the block equal in magnitude and opposite in direction to the block's weight.

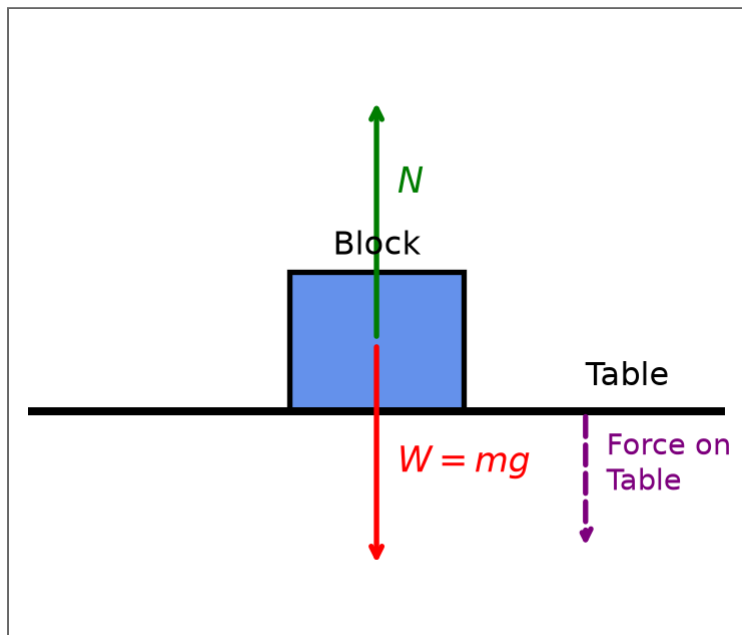


Figure 5: A Mass Resting on a Horizontal Table



Example: Block on a Table

A block of mass 5 kg rests on a horizontal table. Determine the normal force exerted by the table on the block and state the reaction force exerted by the block on the table.

Given $m = 5 \text{ kg}$, $g = 9.8 \text{ m/s}^2$. The forces acting on the block are:

- Weight, $W = mg$, acting vertically downward.
- Normal force, N , exerted by the table, acting vertically upward.

Since the block is at rest, the net vertical force is zero, $\sum F_y = 0$.

$$\therefore N - W = 0 \quad \Rightarrow \quad N = W = mg = (5)(9.8) = 49 \text{ N}.$$

According to Newton's third law, forces always occur in equal and opposite pairs acting on different objects.

- **Action:** The table exerts an upward normal force of 49 N on the block.
- **Reaction:** The block exerts an equal downward force of 49 N on the table.



EXAMPLE: BLOCK ON AN INCLINED PLANE

A block of mass 10 kg rests on a smooth inclined plane that makes an angle of 30° with the horizontal. Determine the normal force exerted by the plane on the block. Hence, state the reaction force exerted by the block on the plane.

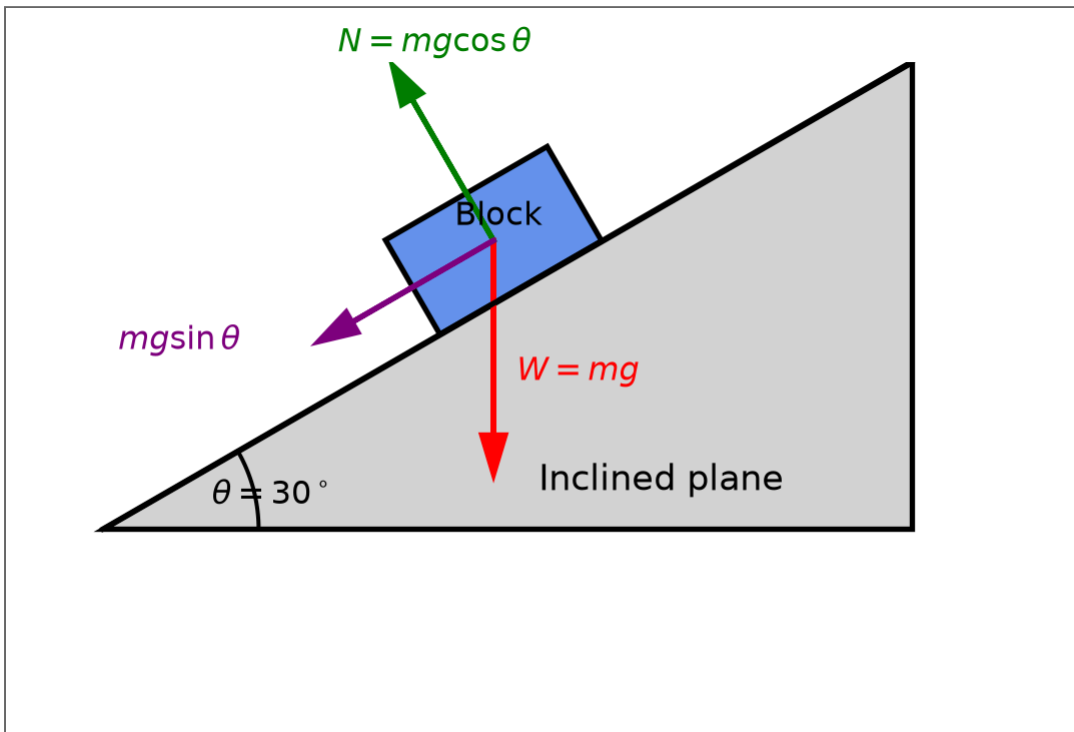


Figure 6: A Block Resting on an Inclined Plane



Given:

$$m = 10 \text{ kg}, \quad g = 9.8 \text{ m/s}^2, \quad \theta = 30^\circ.$$

The weight of the block is

$$W = mg = 10(9.8) = 98 \text{ N}.$$

- The weight acts vertically downward. Since the plane is inclined, resolve the weight into two components:

Perpendicular to the plane:

$$W_{\perp} = mg \cos \theta.$$

Parallel to the plane:

$$W_{\parallel} = mg \sin \theta.$$



Since the block is at rest and the plane is smooth, the normal force is equal to the perpendicular component of the weight:

$$N = mg \cos \theta = 98 \cos 30^\circ = 98(0.866) = 84.9 \text{ N}.$$

According to Newton's third law, every action force has an equal and opposite reaction force.

- **Action:** The inclined plane exerts a normal force of 84.9 N on the block.
- **Reaction:** The block exerts an equal and opposite normal force of 84.9 N on the inclined plane.



EXAMPLE: HANGING MASS

When a mass is suspended from a rope, the mass exerts a downward force on the rope equal to its weight. The rope exerts an equal and opposite upward force on the mass.

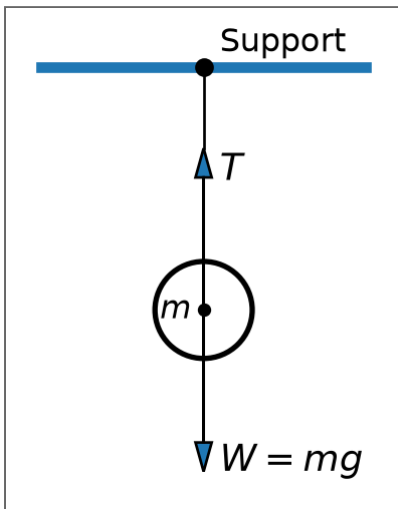


Figure 7: A Mass Suspended by a Light String

Figure 7 illustrates a mass suspended by a light string. The mass exerts a downward force ($W = mg$) on the string equal to its weight, and the string exerts an equal and opposite upward force (T) on the mass.



Other examples

Rocket Propulsion

A rocket expels gas at high speed from its engines. The reaction force exerted by the expelled gas on the rocket is equal in magnitude and opposite in direction to the force exerted by the rocket on the gas.



Swimming

When a swimmer pushes against the water with their hands and feet, the water exerts an equal and opposite force on the swimmer, propelling them forward.



Walking

When a person walks, they push the ground backward with their feet. The ground exerts an equal and opposite force on the person, propelling them forward.



Bird Flight

When a bird flaps its wings, it pushes the air downward and backward. The air pushes back on the bird with an equal and opposite force, allowing the bird to fly.



Boat Sailing

When a boat's sail pushes against the wind, the wind pushes back on the sail with an equal and opposite force.



Car Driving

When a car's tires push backward against the road, the road pushes forward on the tires with an equal and opposite force.



Free-Body Diagram

Free-body diagrams provide the starting point for deriving equations of motion. A free-body diagram should include

- all external forces,
- appropriate coordinate axes,
- force directions,
- known quantities and unknown reactions.



Free-body diagrams are essential for identifying forces before deriving equations of motion.



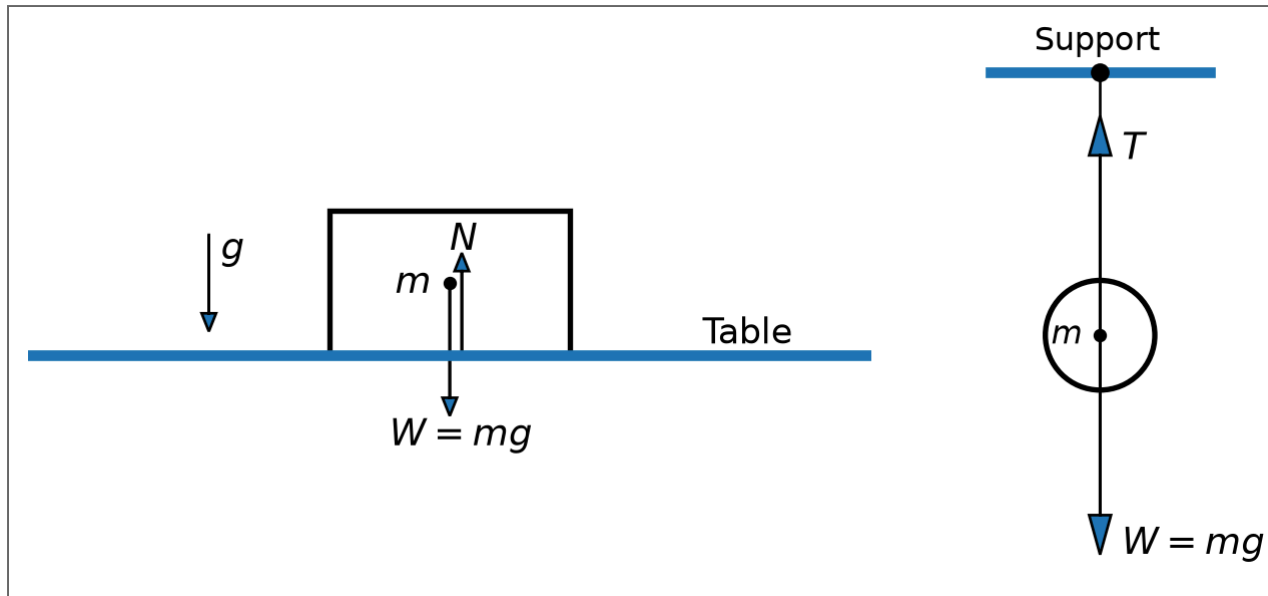


Figure 8: Free-Body Diagram of a Mass

Limitations of Newtonian Mechanics

Although Newton's laws describe many engineering systems with remarkable accuracy, they have limitations.

They are not suitable for

- atomic and subatomic particles, where quantum mechanics is required;
- objects moving at speeds close to the speed of light, where Einstein's theory of relativity applies;
- extremely large astrophysical systems where relativistic effects become significant.

For most engineering applications involving ordinary speeds and dimensions, Newton's laws provide an excellent mathematical model.



Conservation Principles

- Most mathematical models are derived from conservation laws.
- These state that certain physical quantities cannot be created or destroyed, although they may be transported or transformed.

The three fundamental conservation laws are

- Conservation of **Mass**
- Conservation of **Energy**
- Conservation of **Momentum**

Each conservation law has a common mathematical structure,

$$\text{Accumulation} = \text{Input} - \text{Output} + \text{Generation}.$$

This general balance equation provides the starting point for developing many engineering models.



CONSERVATION OF MASS

The principle of conservation of mass states that mass can neither be created nor destroyed.

For a control volume,

$$\text{Rate of accumulation} = \text{Mass inflow} - \text{Mass outflow}.$$

For a closed system,

$$m = \text{constant}.$$

In a steady-state flow through a control volume, the rate of mass accumulation is zero. Therefore, the mass inflow equals the mass outflow.



Example: Conservation of Mass in a Water Tank

Suppose water flows into a tank at a rate of 5 kg/s and flows out at a rate of 3 kg/s. The rate of mass accumulation in the tank is

$$\frac{dm}{dt} = 5 - 3 = 2 \text{ kg/s.}$$

Example: Conservation of Mass in a Pipe

Suppose water flows through a pipe with a cross-sectional area of $A_1 = 0.1 \text{ m}^2$ at a velocity of $v_1 = 2 \text{ m/s}$ and a density of $\rho = 1000 \text{ kg/m}^3$.

The mass flow rate is

$$\dot{m} = \rho A_1 v_1 = 1000 \text{ kg/m}^3 \times 0.1 \text{ m}^2 \times 2 \text{ m/s} = 200 \text{ kg/s.}$$



Example: Bank Account Analogy

Suppose you have a bank account with an initial balance of B_0 . If you deposit money at a rate of $D(t)$ and withdraw money at a rate of $W(t)$, the rate of change of your account balance is given by

$$\frac{dB}{dt} = D(t) - W(t).$$

If you deposit 100 dollars/week and withdraw 50 dollars/week, the rate of change of your account balance is

$$\frac{dB}{dt} = 100 - 50 = 50 \text{ dollars/week.}$$



To find the account balance at any time t , you can integrate this equation over time, taking into account the initial balance $B_0 = 500$.

$$\begin{aligned}\int_0^t \frac{dB}{dt} dt &= \int_0^t (D(t) - W(t)) dt \\ &= \int_0^t 50 dt \\ &= 50t + B_0 \\ B(t) &= 50t + 500.\end{aligned}$$



Example: Conservation of Mass in Traffic Density

Suppose the traffic density on a road segment is $\rho(x, t)$ (vehicles per unit length) and the traffic flow rate is $q(x, t)$ (vehicles per unit time).

The conservation of mass for traffic can be expressed as

$$\frac{\partial \rho}{\partial t} + \frac{\partial q}{\partial x} = 0,$$

- $q = \rho v$ and v is the average speed of vehicles.
- This equation states that the rate of change of traffic density at a point is equal to the negative of the spatial change in traffic flow.



CONSERVATION OF MOMENTUM

The principle of conservation of momentum states that the total momentum of a closed system remains constant if no external forces act on it.

$$\sum \mathbf{P}_{\text{before}} = \sum \mathbf{P}_{\text{after}}$$

- $\mathbf{P}$ is the momentum of the system, and $\mathbf{P} = m\mathbf{v}$ for a particle of mass m and velocity $\mathbf{v}$; therefore, for a system of n particles,

$$\mathbf{P}_1 + \mathbf{P}_2 + \dots + \mathbf{P}_n = \mathbf{P}'_1 + \mathbf{P}'_2 + \dots + \mathbf{P}'_n$$

$$m_1\mathbf{v}_1 + m_2\mathbf{v}_2 + \dots + m_n\mathbf{v}_n = m'_1\mathbf{v}'_1 + m'_2\mathbf{v}'_2 + \dots + m'_n\mathbf{v}'_n$$

Momentum conservation forms the basis of

- rigid-body mechanics,
- fluid mechanics,
- impact analysis,
- vehicle dynamics.



Example: Conservation of Momentum in Collisions

Suppose two ice skaters, Skater A and Skater B, collide on an ice rink. Skater A has a mass of $m_A = 50$ kg and is moving with a velocity of $v_A = 2$ m/s, while Skater B has a mass of $m_B = 70$ kg and is initially at rest. After the collision, they move together with a common velocity v_f .

Using the conservation of momentum:

$$m_A v_A + m_B v_B = (m_A + m_B) v_f$$

Substituting the values:

$$50 \times 2 + 70 \times 0 = (50 + 70) v_f$$

$$100 = 120 v_f$$

$$v_f = \frac{100}{120} \approx 0.833 \text{ m/s}$$



Example: Conservation of Momentum in Billiard Ball Collision

A 0.2 kg billiard ball moving at $v_{1x} = 5 \text{ m/s}$ along the $+X$ axis hits a second identical ball at rest ($v_{2x} = 0 \text{ m/s}$). After the collision, the first ball deflects at an angle of 30° above the $+X$ axis, and the second ball moves at an angle of 60° below the $+X$ axis. Using conservation of momentum, find the final velocities of both balls v'_1 and v'_2 .

We resolve the momentum into X and Y components.

The conservation of momentum in the X direction gives

$$m_1 v_{1x} + m_2 v_{2x} = m_1 v'_1 \cos(30^\circ) + m_2 v'_2 \cos(-60^\circ)$$

$$0.2 \times 5 + 0.2 \times 0 = 0.2 v'_1 \left(\frac{\sqrt{3}}{2} \right) + 0.2 v'_2 \left(\frac{1}{2} \right)$$

$$5 = 0.866 v'_1 + 0.5 v'_2 \quad (\text{Eq. 1})$$



The conservation of momentum in the Y direction gives

$$m_1 v_{1y} + m_2 v_{2y} = m_1 v'_1 \sin(30^\circ) + m_2 v'_2 \sin(-60^\circ)$$

$$0.2 \times 0 + 0.2 \times 0 = 0.2 v'_1 \left(\frac{1}{2} \right) + 0.2 v'_2 \left(-\frac{\sqrt{3}}{2} \right)$$

$$0 = 0.5 v'_1 - 0.866 v'_2$$

$$v'_1 = 1.732 v'_2 \quad (\text{Eq. 2})$$

Substituting Eq. 2 into Eq. 1:

$$5 = 0.866(1.732 v'_2) + 0.5 v'_2$$

$$5 = 2 v'_2$$

$$v'_2 = 2.5 \text{ m/s}$$

Substituting v'_2 back into Eq. 2:

$$v'_1 = 1.732 \times 2.5 \approx 4.33 \text{ m/s}$$



Conservation of Energy

The First Law of Thermodynamics states that energy cannot be created or destroyed. For a control volume,

$$\frac{dE_{\text{system}}}{dt} = \dot{E}_{\text{in}} - \dot{E}_{\text{out}} + \dot{E}_{\text{gen}},$$

- $\frac{dE_{\text{system}}}{dt}$ is the **rate of change of energy stored** within the system,
- $\dot{E}_{\text{in}}$ is the **rate of energy entering** the system,
- $\dot{E}_{\text{out}}$ is the **rate of energy leaving** the system, and
- $\dot{E}_{\text{gen}}$ is the **rate of energy generation** within the system.

For systems with no energy generation, the equation simplifies to

$$\frac{dE_{\text{system}}}{dt} = \dot{E}_{\text{in}} - \dot{E}_{\text{out}}.$$



Example: Conservation of Energy in a Chemical Reactor

A well-insulated chemical reactor receives reactants carrying thermal energy at a rate of 120 kW. The products leave the reactor carrying thermal energy at a rate of 80 kW. An exothermic chemical reaction occurring inside the reactor generates thermal energy at a rate of 50 kW. Determine the rate of change of energy stored in the reactor.

Using the conservation of energy equation,

$$\begin{aligned}\frac{dE_{\text{system}}}{dt} &= \dot{E}_{\text{in}} - \dot{E}_{\text{out}} + \dot{E}_{\text{generated}} \\ &= 120 - 80 + 50 \\ &= 90 \text{ kW}.\end{aligned}$$

- The energy stored in the reactor increases at a rate of 90 kW.



Example: Steady-State Energy Balance for a Gas Turbine

A gas turbine operates continuously at steady state. Air enters the turbine system carrying thermal energy at a rate of $\dot{E}_{\text{in}} = 6.0 \text{ MW}$. Fuel combustion inside the system generates thermal energy at a rate of $\dot{E}_{\text{generated}} = 18.0 \text{ MW}$.

The exhaust gases leave the turbine carrying thermal energy at a rate of $\dot{E}_{\text{exhaust}} = 20.5 \text{ MW}$. The turbine produces shaft work at a rate of $\dot{W}_{\text{shaft}} = 3.0 \text{ MW}$ and loses heat to the surroundings at a rate of $\dot{Q}_{\text{loss}} = 0.5 \text{ MW}$. Verify that the turbine operates at **steady state**.

At steady state,

$$\frac{dE_{\text{system}}}{dt} = 0 \quad \text{or} \quad \dot{E}_{\text{in}} + \dot{E}_{\text{generated}} = \dot{E}_{\text{out}}.$$



The total energy leaving the system is

$$\dot{E}_{\text{out}} = \dot{E}_{\text{exhaust}} + \dot{W}_{\text{shaft}} + \dot{Q}_{\text{loss}} = 24.0 \text{ MW}.$$

Substituting into the energy balance,

$$\begin{aligned} \frac{dE_{\text{system}}}{dt} &= \dot{E}_{\text{in}} - \dot{E}_{\text{out}} + \dot{E}_{\text{generated}} \\ &= 6.0 - 24.0 + 18.0 = 0. \end{aligned}$$

- Since the rate of energy change is zero, the turbine is operating at steady state.

Note: In practice, a continuously operating gas turbine is designed to operate at steady state to ensure no damage occurs to the turbine components due to excessive energy accumulation.



Example: Energy Balance for a Gas Turbine Combustor

- A gas turbine combustor operates at steady state. Air enters the combustor at a mass flow rate of 18 kg/s with a specific enthalpy of 320 kJ/kg. Fuel enters at a mass flow rate of 0.40 kg/s with a heating value of 43 000 kJ/kg. Combustion converts the chemical energy of the fuel into thermal energy within the combustor.
- The combustion products leave the combustor at a mass flow rate of 18.4 kg/s with a specific enthalpy of 1 150 kJ/kg. Heat is lost from the combustor to the surroundings at a rate of 0.80 MW. Neglect changes in kinetic and potential energy. Determine whether the combustor is accumulating energy or operating at steady state.

STEP 1. DETERMINE THE RATE OF ENERGY ENTERING.

The incoming air contributes

$$\dot{E}_{\text{air}} = \dot{m}_{\text{air}} h_{\text{air}} = (18)(320) = 5760 \text{ kW}.$$



STEP 2. DETERMINE THE RATE OF ENERGY LEAVING.

The combustion products carry away

$$\dot{E}_{\text{products}} = (18.4)(1150) = 21160 \text{ kW.}$$

Heat loss to the surroundings is $\dot{Q}_{\text{loss}} = 800 \text{ kW.}$

$$\dot{E}_{\text{out}} = 21160 + 800 = 21960 \text{ kW.}$$

STEP 3. DETERMINE THE ENERGY GENERATED.

The combustion of fuel releases

$$\dot{E}_{\text{generated}} = \dot{m}_{\text{fuel}}(\text{Heating Value}) = (0.40)(43000) = 17200 \text{ kW.}$$



STEP 4. APPLY THE CONSERVATION OF ENERGY EQUATION.

$$\begin{aligned}\frac{dE_{\text{system}}}{dt} &= 5760 - 21960 + 17200 \\ &= 1000 \text{ kW} = 1.0 \text{ MW}.\end{aligned}$$

Since the rate of energy change is positive, the combustor is accumulating energy at a rate of 1.0 MW.

Note: If this condition persisted indefinitely, temperatures and stresses would continue to rise, which could eventually damage the equipment. Therefore, a sustained positive accumulation is not a normal operating condition.



Applications of Newton's Laws - Projectile Motion

By identifying the forces acting on the projectile and applying Newton's Second Law, we obtain a system of ordinary differential equations describing its motion.

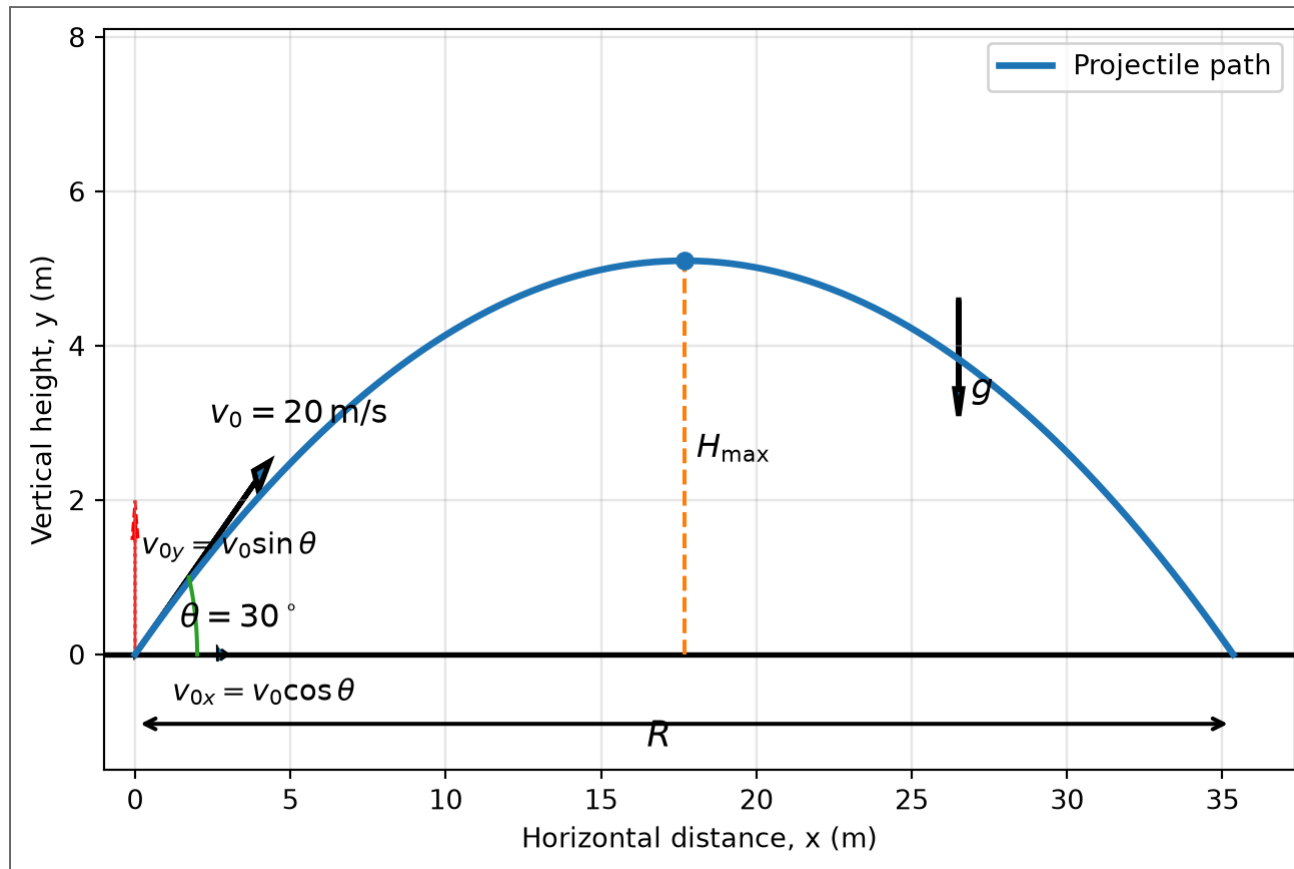


Figure 9: Projectile Motion of a Ball



Example: Projectile Motion of a Ball

A ball is thrown with an initial velocity of $v_0 = 20 \text{ m/s}$ at an angle of $\theta = 30^\circ$ above the horizontal. Determine the trajectory of the ball, its maximum height, and the time it takes to hit the ground.

Given

$$v_0 = 20 \text{ m/s}, \quad \theta = 30^\circ, \quad g = 9.8 \text{ m/s}^2.$$

Projectile Motion Resolve the initial velocity into horizontal and vertical components:

$$v_{0x} = v_0 \cos \theta \quad \text{and} \quad v_{0y} = v_0 \sin \theta$$

Substituting,

$$v_{0x} = 20 \cos 30^\circ = 20(0.866) = 17.32 \text{ m/s}$$

$$v_{0y} = 20 \sin 30^\circ = 20(0.5) = 10 \text{ m/s}.$$



Trajectory of the Ball

The horizontal and vertical positions are

$$x(t) = v_{0x}t \quad \text{and} \quad y(t) = v_{0y}t - \frac{1}{2}gt^2.$$

Thus,

$$x(t) = 17.32t \quad \text{and} \quad y(t) = 10t - 4.9t^2.$$

To write the trajectory as y in terms of x , use

$$t = \frac{x}{17.32}.$$

Substitute into $y(t)$:

$$y = 10 \left(\frac{x}{17.32} \right) - 4.9 \left(\frac{x}{17.32} \right)^2 = 0.577x - 0.0163x^2.$$



Maximum Height

$$H = \frac{v_{0y}^2}{2g}.$$

Substituting,

$$H = \frac{10^2}{2(9.8)} = \frac{100}{19.6} = 5.10 \text{ m.}$$

Time to Hit the Ground

$$y(t) = 10t - 4.9t^2 = 0 \quad \rightarrow \quad t(10 - 4.9t) = 0.$$

Ignoring $t = 0$, the time of flight is

$$t = \frac{10}{4.9} = 2.04 \text{ s.}$$



UNIFORM CIRCULAR MOTION

Let a particle of mass m move in a circle of radius r at a constant speed v .

The particle experiences a centripetal acceleration directed toward the center of the circle, given by

$$a_c = \frac{v^2}{r}.$$

The centripetal force required to keep the particle moving in a circle is

$$F_c = ma_c = m \frac{v^2}{r}.$$



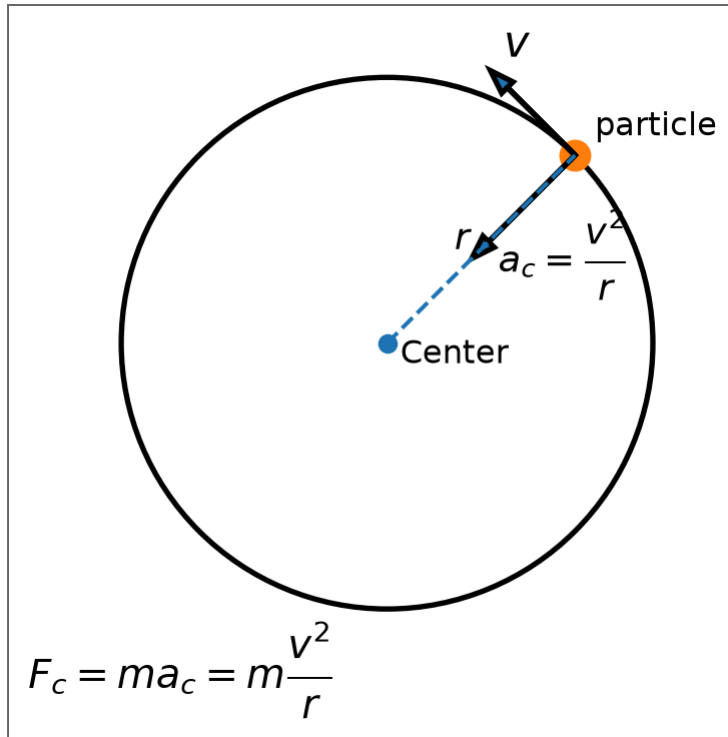


Figure 10: Uniform Circular Motion of a Particle

Example: Car Turning Around a Circular Curve

A car of mass 1200 kg travels around a circular curve of radius 50 m at a constant speed of 15 m/s. Determine the centripetal acceleration and the centripetal force acting on the car.

The centripetal acceleration is

$$a_c = \frac{v^2}{r}.$$

Substituting,

$$a_c = \frac{15^2}{50} = \frac{225}{50} = 4.5 \text{ m/s}^2.$$

The centripetal force is

$$F_c = ma_c.$$

$$\therefore F_c = (1200)(4.5) = 5400 \text{ N}.$$



MOVING IN A CIRCLE WITH FRICTION

- When a vehicle moves in a circular path, friction between the tires and the road provides the necessary centripetal force to keep the vehicle moving in a circle.
- The maximum speed at which a vehicle can safely navigate a curve without skidding is determined by the coefficient of static friction between the tires and the road.

Example: Maximum Speed on a Curved Road

A car of mass 1000 kg is traveling around a circular curve of radius 30 m. The coefficient of static friction between the tires and the road is $\mu_s = 0.6$. Determine the maximum speed at which the car can safely navigate the curve without skidding.

The maximum static friction force that can act on the car is given by

$$f_{\max} = \mu_s N,$$

where N is the normal force.



On a flat road, the normal force is equal to the weight of the car, $N = mg$.

$$\therefore f_{\max} = \mu_s mg.$$

The maximum speed is determined by equating the centripetal force to the maximum static friction force:

$$f_{\max} = \frac{mv^2}{r}.$$

Equating the two expressions for the maximum force, we get

$$\mu_s mg = \frac{mv^2}{r}.$$

Solving for the maximum speed, we get

$$v_{\max} = \sqrt{\mu_s gr} = \sqrt{0.6 \cdot 9.8 \cdot 30} \approx 13.3 \text{ m/s}.$$



OBJECT MOVING IN A ELEVATOR

When an object is moving in an elevator, the forces acting on it include its weight and the normal force exerted by the floor of the elevator. The apparent weight of the object can change depending on the acceleration of the elevator.

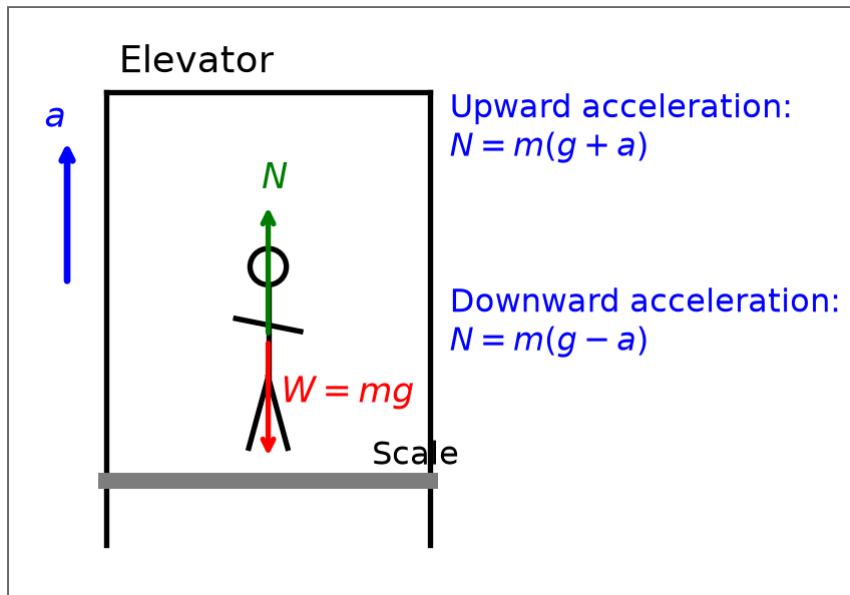


Figure 11: Apparent Weight in an Elevator

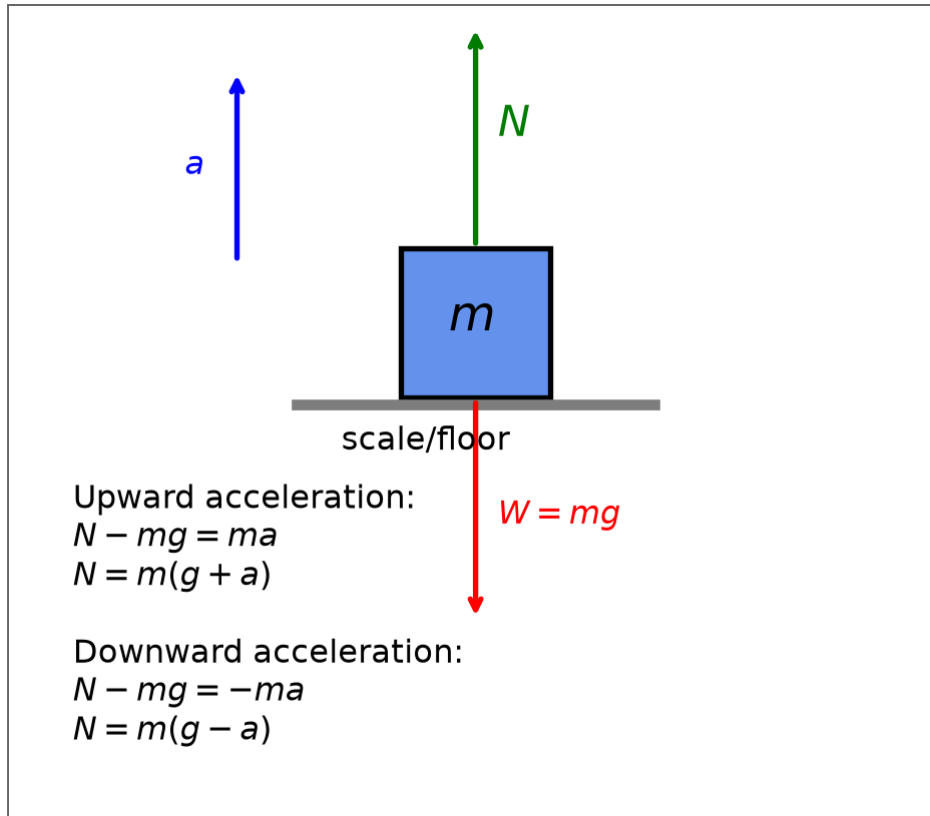


Figure 12: Free-Body Diagram: Apparent Weight in an Elevator

Example: Apparent Weight in an Elevator

A person of mass 70 kg is standing on a scale in an elevator. Determine the reading on the scale (apparent weight) when the elevator is:

1. Moving upward with an acceleration of 2 m/s^2 .
2. Moving downward with an acceleration of 2 m/s^2 .

Given:

$$m = 70 \text{ kg}, \quad g = 9.8 \text{ m/s}^2, \quad a = 2 \text{ m/s}^2.$$

The apparent weight is the normal force N exerted by the scale on the person.



1. Elevator Moving Upward

When the elevator accelerates upward, the forces acting on the person are:

- Normal force, N , acting upward.
- Weight, $W = mg$, acting downward.

Applying Newton's second law,

$$\sum F_y = ma.$$

Taking upward as the positive direction,

$$N - mg = ma.$$

$$\therefore N = m(g + a) = 70(9.8 + 2) = 70(11.8) = 826 \text{ N}.$$



2. Elevator Moving Downward

The forces remain the same, but the acceleration is downward.

Taking upward as the positive direction,

$$N - mg = -ma \quad \rightarrow \quad N = mg - ma.$$

$$N = 70(9.8 - 2) = 70(7.8) = 546 \text{ N}.$$

Elevator Motion	Apparent Weight
Accelerating upward at 2 m/s^2	826 N
Accelerating downward at 2 m/s^2	546 N

- When the elevator accelerates **upward**, the apparent weight **increases** because the scale must provide an additional upward force.
- When the elevator accelerates **downward**, the apparent weight **decreases** because less upward force is required.



SMOOTH PULLEYS MOTION

A mass of m_2 kg is placed on a smooth horizontal table, and connected by a light inextensible string passing over a small smooth pulley at the edge to a mass of m_1 kg hanging freely.

Assume the tension in the string is T . Directions of all forces acting on m_1 is downward, and on m_2 is horizontal. The acceleration of the system is a .

The equation of motion for m_1 by using Newton's 2nd law is

$$m_1 g - T = m_1 a$$

The equation of motion for m_2 by using Newton's 2nd law is

$$T = m_2 a$$

The equations for m_1 and m_2 constitute a system of two equations with two unknowns tension force T and acceleration a .



Solve the system of equations to find the acceleration a and tension T in the string.

$$a = \frac{m_1 g}{m_1 + m_2}, \quad T = \frac{m_1 m_2 g}{m_1 + m_2}.$$

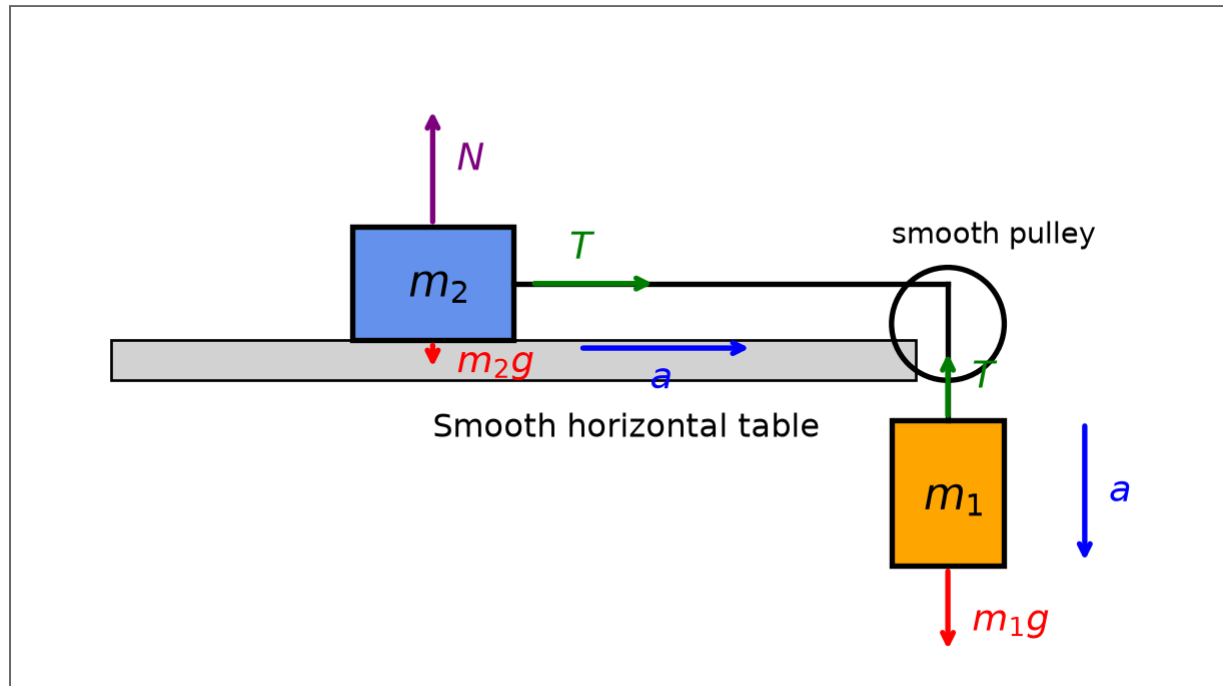


Figure 13: Motion of Two Masses Connected by a Smooth Pulley



Example: Motion of Two Masses Connected by a Smooth Pulley

A mass of $m_1 = 4 \text{ kg}$ hangs freely over the edge of a smooth table and is connected by a light inextensible string passing over a smooth pulley to a mass of $m_2 = 6 \text{ kg}$ resting on the table. Taking $g = 9.8 \text{ m/s}^2$ and assuming the pulley and string are ideal.

Determine:

1. The acceleration of the system.
2. The tension in the string.



STEP 1: DRAW THE FREE-BODY DIAGRAMS

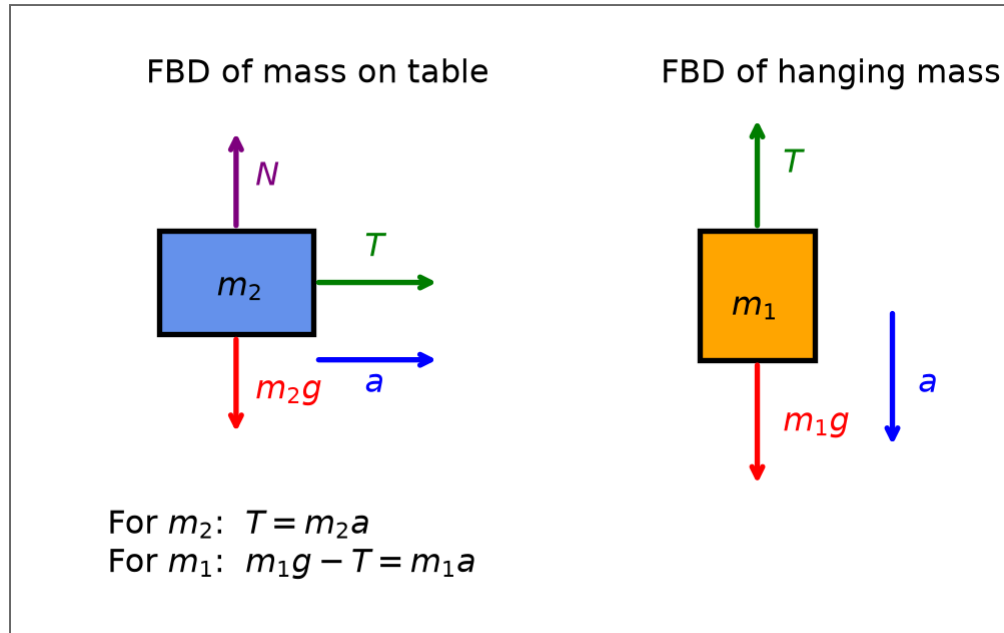


Figure 14: Free-Body Diagrams of the Two Masses

For the hanging mass m_1 :

- Weight: m_1g acting downward.
- Tension: T acting upward.

Since the mass accelerates downward,



$$m_1 g - T = m_1 a.$$



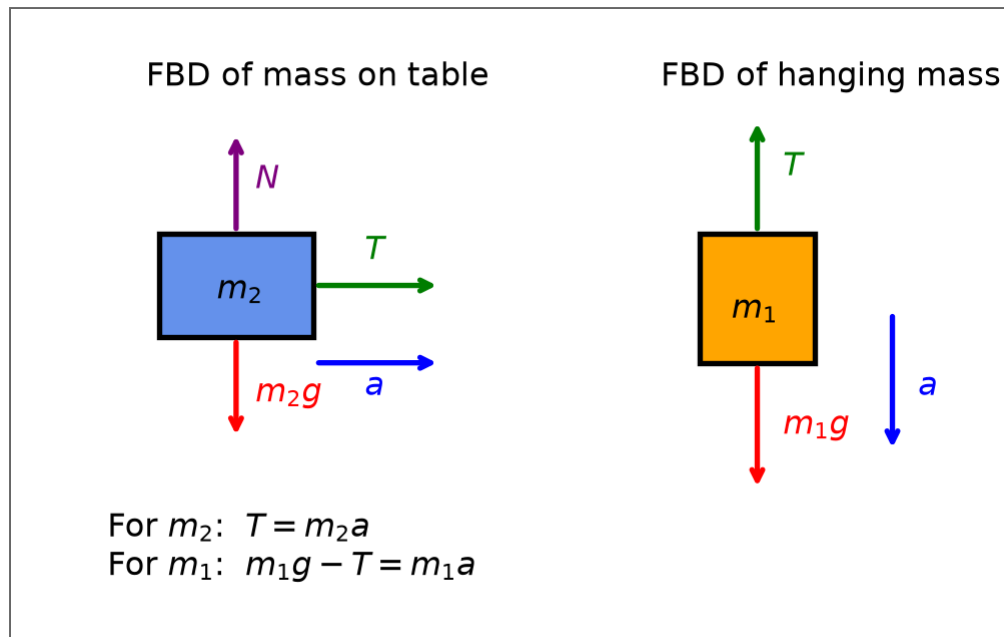


Figure 15: Free-Body Diagrams of the Two Masses

For the mass m_2 on the table:

- Tension: T acting horizontally.
- The table is smooth, so there is no friction.

Applying Newton's second law,

$$T = m_2a.$$



STEP 2: SOLVE FOR THE ACCELERATION

Substitute $T = m_2 a$ into $m_1 g - T = m_1 a$.

then $m_1 g - m_2 a = m_1 a \rightarrow m_1 g = (m_1 + m_2) a$

Substituting the given values yields the acceleration of the system

$$a = \frac{4(9.8)}{4 + 6} = \frac{39.2}{10} = 3.92 \text{ m/s}^2.$$

STEP 3: SOLVE FOR THE TENSION

Using $T = m_2 a$, we obtain the tension in the string

$$T = (6)(3.92) = 23.52 \text{ N}.$$



Verification

Using the formula

$$T = \frac{m_1 m_2 g}{m_1 + m_2},$$

we have

$$T = \frac{(4)(6)(9.8)}{4 + 6} = \frac{235.2}{10} = 23.52 \text{ N},$$

which agrees with the previous result.



INCLINED PLANE WITH FRICTION

When a block slides down an inclined plane, the forces acting on it include its weight, the normal force from the plane, and the frictional force opposing the motion. The component of the weight parallel to the incline causes the block to accelerate down the slope.

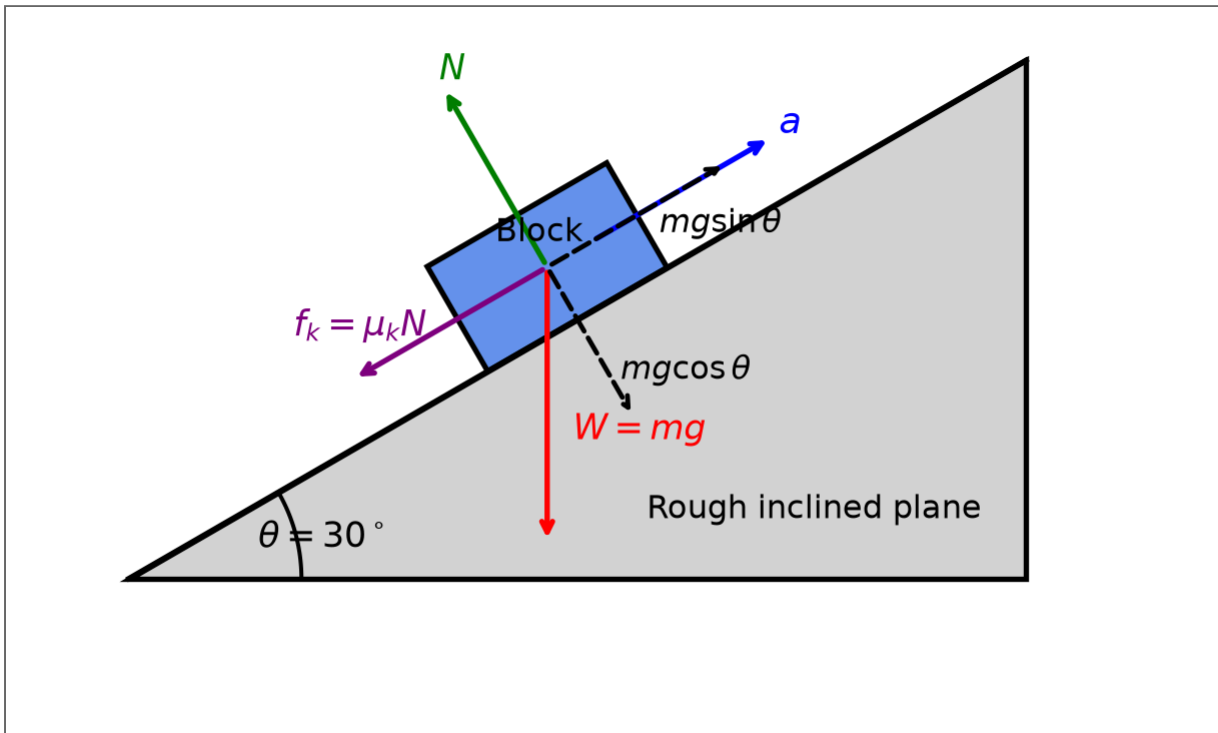


Figure 16: Block Sliding Down an Inclined Plane with Friction



Example: Block Sliding Down an Inclined Plane with Friction

A block of mass 5 kg is placed on a rough inclined plane that makes an angle of 30° with the horizontal. The coefficient of kinetic friction between the block and the plane is $\mu_k = 0.20$. Take $g = 9.8$ and determine:

1. The acceleration of the block as it slides down the incline.
2. The normal force acting on the block.

STEP 1: DRAW THE FREE-BODY DIAGRAM

The forces acting on the block are:

- Weight: $W = mg$, acting vertically downward.
- Normal force: N , acting perpendicular to the plane.
- Frictional force: $f_k = \mu_k N$, acting up the plane, opposite to the motion.



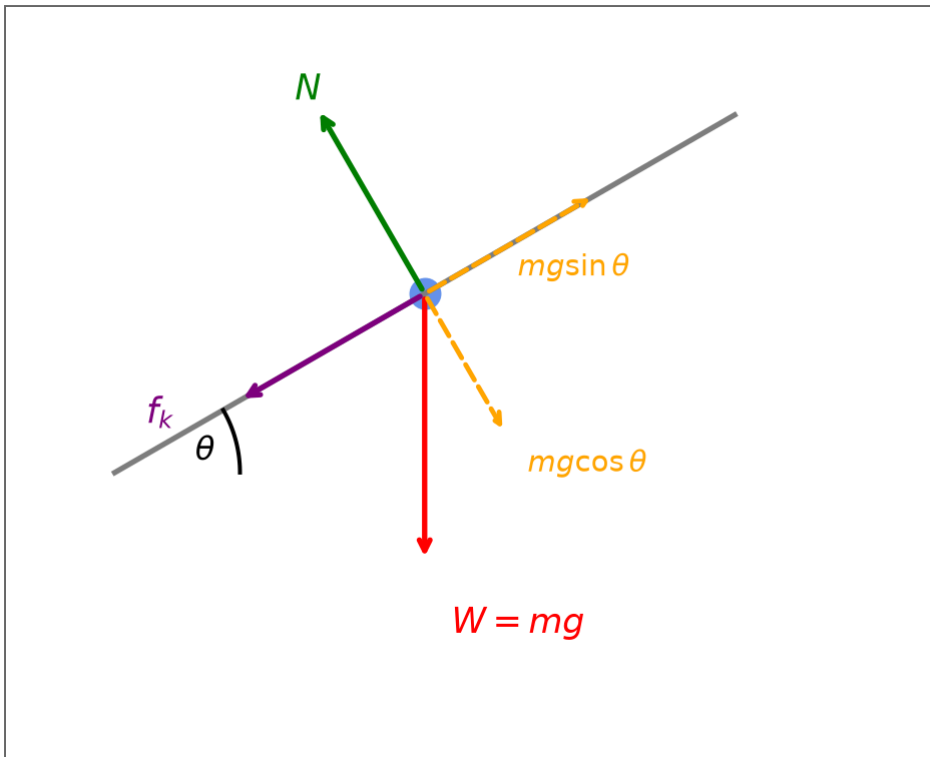


Figure 17: Free-Body Diagram of a Block on an Inclined Plane

Resolve the weight into components:

- Parallel to the incline: $W_{\parallel} = mg \sin \theta$.
- Perpendicular to the incline: $W_{\perp} = mg \cos \theta$.



STEP 2: CALCULATE THE NORMAL FORCE

Since there is no acceleration perpendicular to the plane,
 $N = mg \cos \theta = (5)(9.8) \cos 30^\circ = 49(0.866) = 42.43 \text{ N}.$

STEP 3: CALCULATE THE FRICTIONAL FORCE

Using

$$f_k = \mu_k N,$$

we obtain

$$f_k = (0.20)(42.43) = 8.49 \text{ N}.$$



STEP 4: APPLY NEWTON'S SECOND LAW

The net force along the incline is

$$mg \sin \theta - f_k = ma.$$

Substituting the values,

$$(5)(9.8) \sin 30^\circ - 8.49 = 5a.$$

Since $\sin 30^\circ = 0.5$, we have

$$24.5 - 8.49 = 5a \Rightarrow a = \frac{16.01}{5} = 3.20 \text{ m/s}^2.$$













Speaker notes

